13 Applications of Integration

13.1 Applications of Multiple Integrals

Definition 13.1: Mass in Three Dimensions

The mass of a three-dimensional object with density function $\rho(x, y, z)$ is given by

$$m = \iiint_{\mathcal{W}} \rho(x, y, z)\, \mathrm{d}V$$

where $\mathcal{W}$ is the region occupied by the object.

Definition 13.2: Center of Mass in Three Dimensions

The center of mass $(\bar{x}, \bar{y}, \bar{z})$ of a three-dimensional object with density function $\rho(x, y, z)$ is given by

$$\begin{cases} \bar{x} = \dfrac{1}{m} \iiint_{\mathcal{W}} x\, \rho(x, y, z)\, \mathrm{d}V, \\ \bar{y} = \dfrac{1}{m} \iiint_{\mathcal{W}} y\, \rho(x, y, z)\, \mathrm{d}V, \\ \bar{z} = \dfrac{1}{m} \iiint_{\mathcal{W}} z\, \rho(x, y, z)\, \mathrm{d}V \end{cases}$$

where m is the mass of the object and $\mathcal{W}$ is the region occupied by the object.

Note that the center of mass transforms as the solid transforms. Therefore, we can use symmetry to find the center of mass of a solid.

For example, if a solid is rotating about a line which passes through the center of mass, then the center of mass must be invariant under the rotation.

Problem 13.1: Find the Center of Mass of a Solid

Find the center of mass of a solid bounded by the surfaces $z \geq \sqrt{x^2 + y^2}$ and $2 \leq z \leq 4$. The density function of the solid is given by $\rho(x, y, z) = 10 - z$.

Solution 13.1

First, we need to find the mass of the solid. The mass is given by

$$m = \iiint_{\mathcal{W}} \rho(x, y, z)\,\mathrm{d}V = \iiint_{\mathcal{W}} 10 - z\,\mathrm{d}V.$$

Note that the solid is rotationally symmetric about the z-axis. Therefore, it might be easier to use cylindrical coordinates to evaluate the integral. In cylindrical coordinates, we have $x = r\cos\theta$, $y = r\sin\theta$, and $z = z$.

Hence, the mass of the solid can be expressed as

$$\begin{aligned}
m &= \int_{z=2}^{4}\int_{\theta=0}^{2\pi}\int_{r=0}^{z}(10-z)r\,\mathrm{d}r\,\mathrm{d}\theta\,\mathrm{d}z \\
&= \int_{z=2}^{4}\int_{\theta=0}^{2\pi}\left[\int_{r=0}^{z}(10-z)r\,\mathrm{d}r\right]\mathrm{d}\theta\,\mathrm{d}z \\
&= \int_{z=2}^{4}\int_{\theta=0}^{2\pi}(10-z)\left[\frac{r^2}{2}\right]_{r=0}^{r=z}\mathrm{d}\theta\,\mathrm{d}z \\
&= \int_{z=2}^{4}\int_{\theta=0}^{2\pi}(10-z)\cdot\frac{z^2}{2}\,\mathrm{d}\theta\,\mathrm{d}z \\
&= \int_{z=2}^{4}(10-z)\cdot\frac{z^2}{2}\cdot 2\pi\,\mathrm{d}z \\
&= \pi\int_{z=2}^{4}(10-z)z^2\,dz \\
&= \pi\left[10\frac{z^3}{3} - \frac{z^4}{4}\right]_{z=2}^{z=4} \\
&= \pi\left(\frac{640}{3} - \frac{256}{4} - \frac{80}{3} + 4\right) \\
&= \frac{380}{3}\pi
\end{aligned}$$

By symmetry, we know that $\bar{x} = 0$ and $\bar{y} = 0$. Next, we need to find $\bar{z}$. The z-

coordinate of the center of mass is given by

$$\begin{aligned}\bar{z} &= \frac{1}{m}\iiint_{\mathcal{W}} z(10-z)\,\mathrm{d}V \\ &= \cdots = \frac{1506}{475}\end{aligned}$$

Hence, the center of mass of the solid is given by

$$(\bar{x}, \bar{y}, \bar{z}) = \left(0, 0, \frac{1506}{475}\right).$$

13.2 Probability

For a discrete random variable, the probability is given by

$$\mathbb{P}(\text{an event}) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}.$$

Similarly, for a uniform continuous random variable, the "total number of outcomes" is given by the area of the region $\mathcal{D}$ in which the random variable is defined. Note that $\mathcal{D} \subseteq \mathbb{R}^n$ for some $n \geq 1$.

If the random variable is not uniform, however, we need to use a density function p to describe the distribution of the random variable.

Definition 13.3: Joint Probability Density Function

A joint probability density function for two continuous random variables X and Y is a function $p(x, y)$ such that

- $p(x, y) \geq 0$ for all $(x, y) \in \mathbb{R}^2$,
- $\displaystyle\iint_{\mathbb{R}^2} p(x, y)\,\mathrm{d}A = 1$

For a region $\mathcal{D} \subseteq \mathbb{R}^2$, the probability that (X, Y) lies in $\mathcal{D}$ is given by

$$\frac{\displaystyle\iint_{\mathcal{D}} p(x,y)\,\mathrm{d}A}{\displaystyle\iint_{\mathbb{R}^2} p(x,y)\,\mathrm{d}A} = \iint_{\mathcal{D}} p(x,y)\,\mathrm{d}A.$$